

3–4 COMPACT PDU SERVICE

3–4–1 Service a Circuit Board



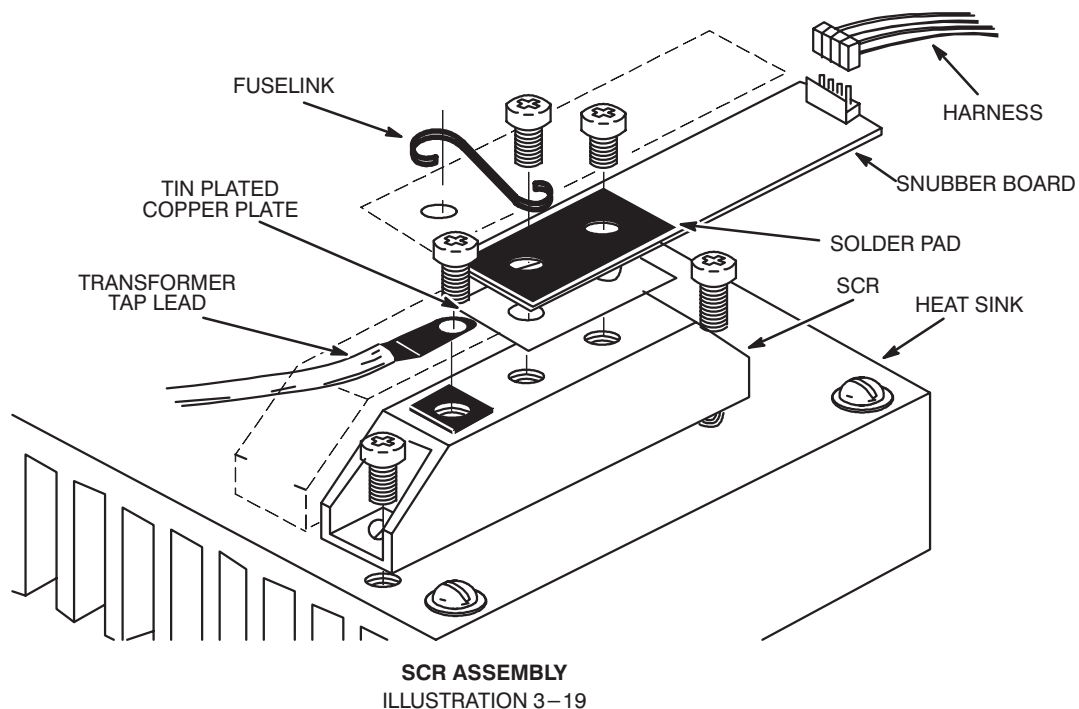
LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Compact PDU.
3. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
4. Turn main circuit breaker OFF.
5. Turn load panel circuit breakers OFF.
6. Measure input test points (A8) for) VAC before proceeding.
7. Disconnect connectors on circuit board. Remove board from standoffs.
8. Install replacement board on standoffs. Reconnect connectors.
9. If a logic board is replaced, perform prestartup and basic check on logic board. Refer to Sections 3–3–1, Prestartup, and 3–3–2, Basic Check.
10. Perform functional check. Refer to Section 3–3–3, Functional Check.
11. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
12. Unlock facility circuit breaker and turn ON.
13. Turn main circuit breaker ON.
14. Press Emer. Stop Reset switch.
15. Turn load panel circuit breakers ON.
16. Close outer door.

3-4-2 Replace an SCR

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Compact PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points (A8) for) VAC before proceeding.
5. Turn all load panel circuit breakers OFF.
6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Disconnect transformer tap lead from SCR. See Illustration 3-19.



8. Remove fuselinks and snubber board from SCR. Remove SCR from heat sink.

3–4–2 Replace an SCR (continued)

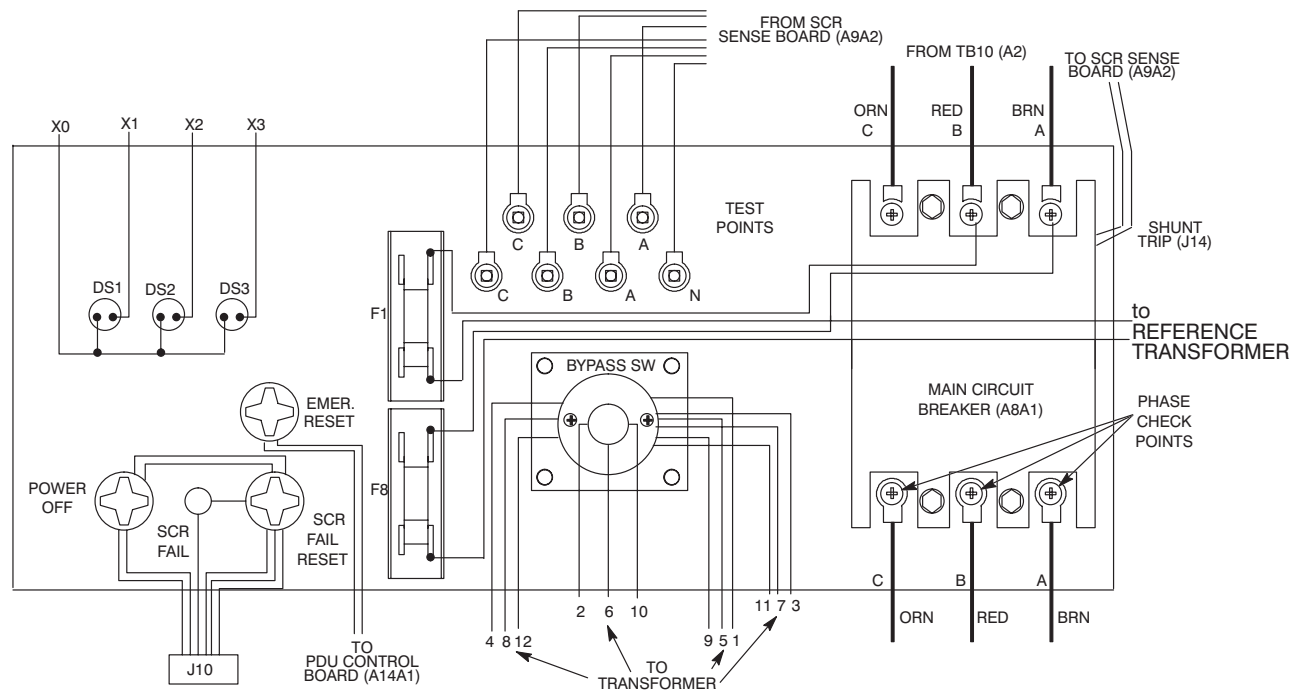
9. Spread a thin film of heat–sink compound on new SCR and mount on heat sink.
10. Attach snubber board to SCR with copper plate under the snubber PCB. Attach fuselinks to snubber board. Torque SCR screws to 30 lb in (34 Nm).
11. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center of right edge of door.
12. Perform basic check. Refer to Section 3–3–2, Basic Check.
13. Perform functional check. Refer to Section 3–3–3, Functional Check.
14. Close outer door.

3–4–3 Replace Main Circuit Breaker (A8A1)

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Compact PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points (A8) for) VAC before proceeding.
5. Turn all load panel circuit breakers OFF.
6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Remove main circuit breaker.
 - a. Disconnect main circuit breaker shunt trip (J14). See Illustration 3–20.
 - b. Remove two small wires at top of main circuit breaker.
 - c. From front panel side, remove and keep four screws, eight washers and four nuts securing main circuit breaker to front panel and 2 spacer plates.
 - d. Remove six power connections to main circuit breaker. Remove main circuit breaker.

3-4-3 Replace Main Circuit Breaker (A8A1) (continued)



FRONT PANEL (REAR VIEW)
ILLUSTRATION 3-20

8. Replace main circuit breaker.

Note

Be sure wire color on input is the same as wire color on output!

- a. Install six power connections to main circuit breaker. Be sure ring terminals are in place on top red and brown power connections. Torque to 25 lb in (113 Nm).
 - b. Holding circuit breaker in place, install four screws, eight washers and four nuts securing main circuit breaker to front panel. Use the two spacer plates.
 - c. Install two wires at top of main circuit breaker.
 - d. Connect main circuit breaker shunt trip (J14).
9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
 10. Unlock facility circuit breaker and turn ON.
 11. Turn main circuit breaker ON.

3–4–3 Replace Main Circuit Breaker (A8A1) (continued)

12. Press emer. stop reset switch.
13. Turn load panel circuit breakers ON.
14. Close outer door.

3–4–4 Replace Bypass Switch (A8A2)

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Compact PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points (A8) for) VAC before proceeding.
5. Turn all load panel circuit breakers OFF.
6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Remove bypass switch. See illustration 3–20.
 - a. On front panel, remove and keep screw securing knob to bypass switch. Remove knob.
 - b. Remove and keep four screws securing bypass switch to front panel. Bypass switch will come free of panel.
 - c. Tag wires on bypass switch to identify where they go on the new bypass switch.
 - d. Remove wires from bypass switch.
 - e. Remove bypass switch.
8. Replace bypass switch.

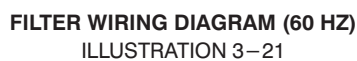


Connect wires to the same location on bypass switch from which they were disconnected. Severe damage to equipment may result if wiring is incorrect.

- a. Connect wires to bypass switch. Torque nine power connections to 30 lb in (34 Nm).

3–4–4 Replace Bypass Switch (A8A2) (continued)

- b. Holding bypass switch in place, install four screws securing bypass switch to front panel.
 - c. Holding knob in place, install screw securing knob to bypass switch.
9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
 10. Unlock facility circuit breaker and turn ON.
 11. Turn main circuit breaker ON.
 12. Press Emer. Stop Reset switch.
 13. Turn load panel circuit breakers ON.
 14. Close outer door.



3–4–5 Replace Surge Suppressor (A11A1)

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock and tag.
2. Open outer door of Compact PDU.
3. Turn off main input circuit breaker.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
6. Remove and keep four nuts, phillips screws, star washers and flat washers securing surge suppressor mount to filter assembly. See Illustration 3–10.
7. Remove surge suppressor assembly.
8. Install new surge suppressor assembly. Secure to filter assembly with screws, star washers and flat washers kept in Step 6.

Note

Be sure wires get to the correct locations: brown to phase A, red to phase B, orange to phase C, and white or blue to neutral on both line and load sides. See Illustration 3–21.

9. Connect all output wires from surge suppressor.
10. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
11. Unlock facility circuit breaker and turn ON.
12. Turn main circuit breaker ON.
13. Press Emer. Stop Reset switch.
14. Turn load panel circuit breakers ON.
15. Close outer door.

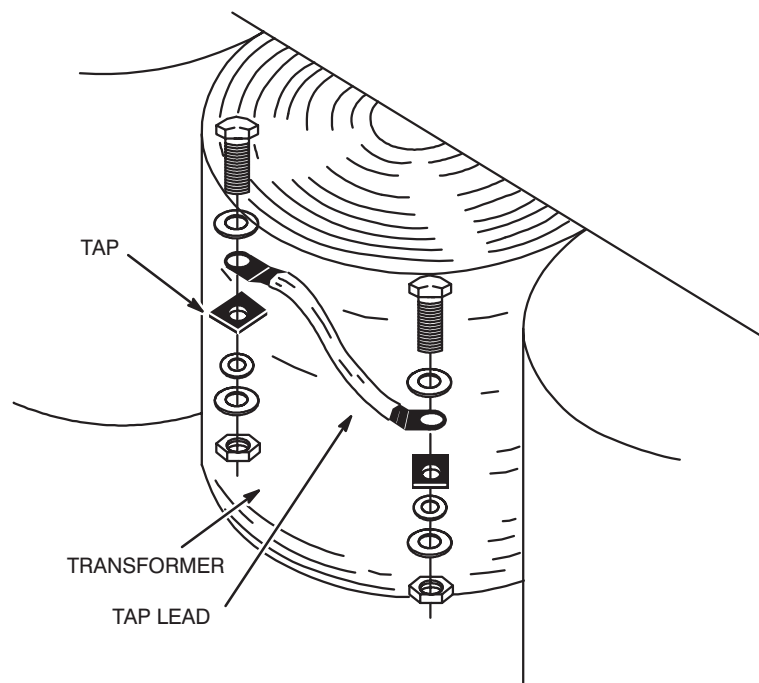
3-4-6 Change Transformer Tap-lead Position (A6T1)

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock and tag.
2. Open outer door of Compact PDU.
3. Measure input test points (A8) for 0 VAC before proceeding.
4. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
5. Unbolt end of tap lead. See Illustration 3-22.
6. Move tap lead to desired tap. Refer to Section 3-2, CONFIGURATION.

Note

There must be a flat washer on each side of a terminal lug. This is a UL requirement.



TRANSFORMER TAP CONNECTION

ILLUSTRATION 3-22

3–4–6 Change Transformer Tap–lead Position (A6T1) (continued)

7. Bolt tap lead. Torque to 60 lb in (68 Nm).
8. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
9. Unlock facility circuit breaker and turn ON.
10. Press Emer. Stop Reset switch.
11. Turn main circuit breaker ON.
12. Turn load panel circuit breakers ON.
13. Close outer door.